

$$x^3 - x^2 + 3x - 10 = 0 \Rightarrow x = 2, x = -\frac{1}{2} - \sqrt{19}i/2, x = -\frac{1}{2} + \sqrt{19}i/2,$$

$$\int_0^x t^2 dt = \frac{x^3}{3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The solutions of the equation  $x^2 - 5x + 4 = 0$  are  $x = 1$  and  $x = 4$ .  
The integral is equal to

$$\int \log (x) d x=x \log (x)-x$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}=\frac{\pi}{\tanh (\pi)}$$

$$x^3-4x^2+5x-2=0\Rightarrow x=1,x=2$$

$$\int_x^{x^3} \sin t \, \mathrm{d} t = \cos (x) - \cos \left(x^3\right)$$

$$\int_{-1}^1 \log \left(x^2+1\right) \mathrm{d} x=-4+2 \log (2)+\pi$$

$$\begin{aligned}\iint \log (x y) d x d y &= \int x \log (x y)-x d y \\ &= x y \log (x y)-2 x y\end{aligned}$$

Inline:  $\arcsin\left(\frac{x}{2}\right) - \frac{\pi}{3} = 0$  Equation:

$$\sin^{-1}\left(\frac{x}{2}\right) - \frac{\pi}{3} = 0 \Rightarrow x = \sqrt{3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The solution of the differential equation

$$2f(x)-3\frac{d}{dx}f(x)+\frac{d^2}{dx^2}f(x)=\cos(x)$$

is

$$f(x)=C_1e^x+C_2e^{2x}-\frac{3}{10}\sin(x)+\frac{1}{10}\cos(x)$$

$$\int_{-1}^1 \tan (x) \mathrm{d} x=-\frac{1}{2} \log \left(\sin ^2(x)-1\right)$$

$$\int_1^2 x \log (x)-x \mathrm{d} x=-\frac{9}{4}+2 \log (2)$$

$$3^3=27$$

$$\begin{cases} 2x-5y=-1 \\ 3x+4y=10 \end{cases} \Rightarrow \quad x=2 \quad \text{and} \quad y=1$$

$$\sum_{k=1}^{10} \frac{1}{n^2} = \frac{1968329}{1270080}$$

$$x^2-4x+3=0\Rightarrow x=3$$

$$\int_{-2}^2 \frac{x^2}{x^2+1} \, \mathrm{d} x = \left. x - \tan^{-1}(x) \right|_{-2}^2 = -2 \tan^{-1}(2) + 4$$

$$\int_{-1}^1 x^2 \mathrm{d} x = \frac{x^3}{3}$$

$$\cos(2x)=1\Rightarrow x=\frac{\pi}{4}\quad\text{and}\quad x=\frac{3\pi}{4}$$

$$\det M = \det \begin{pmatrix} 1 & 2 \\ -3 & -1 \end{pmatrix} = 5$$

$$\begin{pmatrix} \frac{1}{5} \\ -\frac{1}{10} \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -3 \\ 2 & 4 \\ 0 & 1 \end{pmatrix}, M^2 = \begin{pmatrix} 1 & 2 \\ -3 & -1 \end{pmatrix}^2 = \begin{pmatrix} -5 & 0 \\ 0 & -5 \end{pmatrix}$$

$$\int_{-1}^1 \frac{x}{x^2-1} \mathrm{d} x =$$

$$\sum_{k=1}^\infty \begin{pmatrix} -5 & 10 & 2 \\ 0 & -10 & -1 \end{pmatrix}$$